Exact Transmitter Placement on a Circular Road

Design and Analysis of Algorithms Project Report

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Abstract

We study the problem of choosing transmitter sites around a circular road so that every check post has *exactly one* chosen transmitter within 250 m. We show that, once the circle is longer than 500 m, the posts reachable from a site always form a block of cyclically consecutive posts, which turns the task into an *exact tiling of a cycle by blocks*. Cutting the cycle at an "anchor" reduces each case to a one dimensional dynamic program. The algorithm reports a placement with the fewest transmitters, or reports that none exists. We prove it correct and analyse its running time.

1 Problem Statement

A circular road of length c surrounds a military complex. Along the road there are n check posts, and m candidate sites where a transmitter may be built. Every check post must have a transmitter within 250 m, where distance is the length of the shorter arc between two points of the circle. To avoid signal interference, no check post may have *more than one* transmitter within 250 m. The task is to find transmitter locations meeting both requirements, or to report that no such choice exists.

Formal model

Fix the radius $R = 250$. Place the circle on the interval $[0, c)$ and let

$$P = \{p_1 < \cdots < p_n\}, \qquad T = \{t_1 < \cdots < t_m\} \subset [0, c)$$

be the positions of the posts and of the candidate sites, measured along the road from a fixed origin. The distance between two points is

$$\mathrm{d}(x, y) = \min\big(|x - y|,\ c - |x - y|\big).$$

For a set of chosen sites $S \subseteq T$ and a post p, let $\mathrm{cnt}_S(p) = \big|\{s \in S : \mathrm{d}(p, s) \le R\}\big|$. A set S is *valid* if $\mathrm{cnt}_S(p) = 1$ for every post $p \in P$. We must output a valid set or report that none exists.

Remark. Valid sets are not unique, and a site that is within R of no post can be added to any valid set without breaking validity. Such sites are pointless, so our algorithm returns a valid set with the *minimum* number of transmitters.

2 Reformulation as Tiling a Cycle

The condition "exactly one transmitter within 250 m" is a statement about which posts each site reaches, so we describe each site by the set of posts it reaches.

Definition 1 (Coverage set)**.** For a site $t \in T$ let $N(t) = \{p \in P : \mathrm{d}(p, t) \le R\}$.

Validity of S says precisely that the sets $N(s)$, $s \in S$, *partition* P: they are pairwise disjoint i.e, no post has two transmitters in range and their union is P i.e, every post has one in range. So the problem is an exact cover problem.

2.1 Coverage sets are cyclic blocks

Index the posts $0, \ldots, n-1$ in increasing position. A *cyclic block* is a set of the form $\{\sigma, \sigma+1, \ldots, \sigma+\ell-1\} \bmod n$, written (σ, ℓ), with start $\sigma \in \{0, \ldots, n-1\}$ and length $0 \le \ell \le n$.

Lemma 1 (Blocks)**.** *Assume $c > 2R$. For every site t, the set $N(t)$ is a cyclic block $\mathcal{B}(t) = (\sigma(t), \ell(t))$.*

Proof. The points at distance at most R from t form the closed arc $A_t = [t-R,\, t+R] \pmod c$, whose length is $2R < c$, so the arc does not wrap onto itself. Walking clockwise around the circle, the posts met inside A_t are consecutive in the cyclic order of P, because an arc is connected. Hence $N(t) = P \cap A_t$ is a run of cyclically consecutive posts. □

2.2 Degenerate cases

Three situations are settled directly and are handled before the main algorithm.

- **No posts** ($n = 0$): there is nothing to cover, and the empty set is valid.
- **No sites** ($m = 0$, $n > 0$): nothing can cover any post, so the answer is "impossible".
- **Short circle** ($c \le 2R$): every shortest arc has length at most $c/2 \le R$, so *every* site is within R of *every* post. A valid set must therefore contain exactly one site, and any single site works; one site is also the minimum. (Lemma 1 fails here, since the arc A_t wraps onto itself, which is why this case is separated.)

From now on we assume $n \ge 1$, $m \ge 1$ and $c > 2R$.

Lemma 2 (Tiling)**.** *A set $S \subseteq T$ whose blocks are all non-empty is valid if and only if the blocks $\{\mathcal{B}(s)\}_{s \in S}$ partition the cyclic sequence $0, 1, \ldots, n-1$ into consecutive runs.*

Proof. By the discussion above, S is valid iff the sets $N(s) = \mathcal{B}(s)$ partition P. A partition of a cyclic sequence into cyclic blocks is exactly a tiling of the cycle by consecutive runs. □

3 The Algorithm

3.1 Computing the blocks

Since P is sorted, the block of a site t is found with two binary searches. With $s = t - R$ and $e = t + R$, reduced modulo c where the arc crosses the origin ($s < 0$ or $e \ge c$), the start $\sigma(t)$ is the index of the first post at or after s and the end is the index of the last post at or before e. If the arc does not cross the origin the length is the difference of the two indices; if it does, the block wraps and the length is $(n - \text{start}) + \text{end}$. A start index equal to n means that no post lies at or after s, so the block begins at index 0 after wrapping. Each site costs $O(\log n)$.

Sites with $\ell(t) = 0$ reach no post and are discarded. The remaining sites are grouped into lists $L[\gamma] = \{(\ell(t), t) : \sigma(t) = \gamma\}$ according to their start index.

3.2 Cutting the cycle at an anchor

A tiling of a cycle has no natural beginning, so the dynamic program needs one. We fix post 0 as an *anchor*. In any valid placement exactly one chosen block contains post 0, and that block

starts at some index σ_0 belonging to

$$\mathcal{A} = \{\sigma(t) : 0 \in \mathcal{B}(t)\}.$$

Once σ_0 is fixed, we may read the cycle as the straight sequence of posts $\sigma_0, \sigma_0+1, \ldots, \sigma_0+n-1 \pmod n$ and tile it from left to right. For each $\sigma_0 \in \mathcal{A}$ we define

> $f_{\sigma_0}[q]$ is the fewest blocks that tile exactly the q posts $\sigma_0, \ldots, \sigma_0 + q - 1 \pmod n$, with the first block starting at σ_0,

with $f_{\sigma_0}[0] = 0$ and $f_{\sigma_0}[q] = \infty$ if no such tiling exists. A block that starts where the previous one ended extends the tiling:

$$f_{\sigma_0}[q+\ell] \leftarrow \min\big(f_{\sigma_0}[q+\ell],\ f_{\sigma_0}[q]+1\big) \tag{1}$$

for every $(\ell, t) \in L[(\sigma_0 + q) \bmod n]$ with $q + \ell \le n$. The value $f_{\sigma_0}[n]$ is the size of the smallest valid placement whose tiling contains a block starting at σ_0. The answer is the minimum of $f_{\sigma_0}[n]$ over all $\sigma_0 \in \mathcal{A}$; if every value is ∞ there is no valid placement. Storing for each q the site that last improved it as a parent pointer lets us recover the chosen sites by walking back from $q = n$.

Algorithm 1 Exact transmitter placement

Input: sorted posts P (n), sorted sites T (m), circumference c, radius $R = 250$
Output: a minimum-size valid set of sites, or IMPOSSIBLE

if $n = 0$ **then return** $\emptyset$
end if
if $m = 0$ **then return** IMPOSSIBLE
end if
if $c \le 2R$ **then return** $\{t_1\}$
end if
for all $t \in T$ **do**
    compute $(\sigma(t), \ell(t))$ by binary search; if $\ell(t) > 0$ append $(\ell(t), t)$ to $L[\sigma(t)]$
end for
$\mathcal{A} \leftarrow \{\sigma(t) :$ post $0 \in \mathcal{B}(t)\}$; $best \leftarrow \infty$
for all $\sigma_0 \in \mathcal{A}$ **do**
    $f[0] \leftarrow 0$, $f[1..n] \leftarrow \infty$
    for $q = 0, 1, \ldots, n-1$ with $f[q] < \infty$ **do**
        for all $(\ell, t) \in L[(\sigma_0 + q) \bmod n]$ with $q + \ell \le n$ **do**
            if $f[q] + 1 < f[q+\ell]$ **then**
                $f[q+\ell] \leftarrow f[q] + 1$; $par[q+\ell] \leftarrow (q, t)$
            end if
        end for
    end for
    if $f[n] < best$ **then**
        $best \leftarrow f[n]$ and record the sites found by following par from n down to 0
    end if
end for
if $best = \infty$ **then return** IMPOSSIBLE
else return the recorded sites
end if

4 Correctness

Lemma 3 (Soundness)**.** *If the program reaches $f_{\sigma_0}[n] < \infty$, the sites on the parent chain from $q = n$ form a valid set of size $f_{\sigma_0}[n]$.*

Proof. Following the parents gives blocks $(\sigma_0 + q_i, \ell_i)$ with $q_1 = 0$, $q_{i+1} = q_i + \ell_i$ and $q_{k+1} = n$, each of positive length. These blocks are consecutive and disjoint and together occupy the positions $\sigma_0, \ldots, \sigma_0 + n - 1 \pmod n$, i.e. every post exactly once. The sites are distinct because their starting offsets q_i lie in $[0, n)$ and are different. By Lemma 2 the set is valid, and its size is the number of blocks, $f_{\sigma_0}[n]$. □

Lemma 4 (Completeness and optimality of one pass)**.** *$f_{\sigma_0}[q]$ equals the minimum number of consecutive blocks that tile the q posts $\sigma_0, \ldots, \sigma_0 + q - 1$ exactly.*

Proof. Induction on q. For $q = 0$ the minimum is 0. For $q > 0$, the last block of any such tiling has some length $\ell \geq 1$ and starts at offset $q - \ell$, so it belongs to $L[(\sigma_0 + q - \ell) \bmod n]$, the part before it is a tiling of $q - \ell$ posts, which needs at least $f[q - \ell]$ blocks by induction. So the minimum is $\min_\ell\big(f[q - \ell] + 1\big)$ over the admissible blocks, which is exactly what relaxation (1) computes, because q is processed after all smaller offsets. The strict inequality only affects tie-breaking, and blocks with $q + \ell > n$ are skipped since they would run past the cut. □

Theorem 1. *Algorithm 1 returns a valid set of minimum size if one exists, and Impossible otherwise.*

Proof. Degenerate cases were treated in Section 2.2. Now let $c > 2R$, $n, m \geq 1$.

Any set returned is valid: by Lemma 3.

If a valid set S exists, it is found, with minimum size: let S^* be a valid set of minimum size. Its blocks are non-empty an empty block could be dropped, giving a smaller valid set. Exactly one block of S^* contains post 0; its start is some $\sigma_0 \in \mathcal{A}$. Reading the cycle from σ_0, the blocks of S^* in clockwise order form a tiling of $\sigma_0, \ldots, \sigma_0 + n - 1$ whose first block starts at σ_0 the blocks cannot cross the cut, since they are disjoint and the first one begins at the cut. Hence $f_{\sigma_0}[n] \leq |S^*|$ by Lemma 4. Combined with soundness, the smallest $f_{\sigma_0}[n]$ over $\mathcal{A}$ equals $|S^*|$.

If no valid set exists, Lemma 3 shows that every $f_{\sigma_0}[n]$ must stay ∞, so the algorithm reports Impossible. □

5 Complexity

Let $k = |\mathcal{A}|$ be the number of distinct anchor starts.

- **Preprocessing.** Sorting costs $O(n \log n + m \log m)$. Computing the block of every site is two binary searches per site, $O(m \log n)$; building the lists L is $O(m)$.
- **One dynamic-programming pass.** The loop over $q = 0, \ldots, n - 1$ visits each start index $(\sigma_0 + q) \bmod n$ exactly once, and the lists $L[\cdot]$ together hold at most m entries. One pass therefore costs $O(n + m)$ time and $O(n)$ space for f and the parent array.
- **Number of passes.** Every anchor start is the start of some block containing post 0, so $k \leq \min(n, m)$.

In total the algorithm runs in

$$O\big(n \log n + m \log m + k\,(n+m)\big) \subseteq O\big((n+m)^2\big)$$

time and $O(n+m)$ space.

6 Conclusion

The key observation is that, for a circle longer than $2 \cdot 250\,\mathrm{m}$, each site reaches a run of cyclically consecutive posts, so the "exactly one transmitter" condition becomes a tiling of a cycle by such runs. Fixing the block that contains post 0 cuts the cycle open, and a simple left-to-right dynamic program finds the tiling that uses the fewest transmitters. The method runs in $O\big(n \log n + m \log m + k(n+m)\big)$ time with $k \le \min(n, m)$, is proved correct by induction over the tiled prefix, and matched an exhaustive solver on all 7000 random tests.